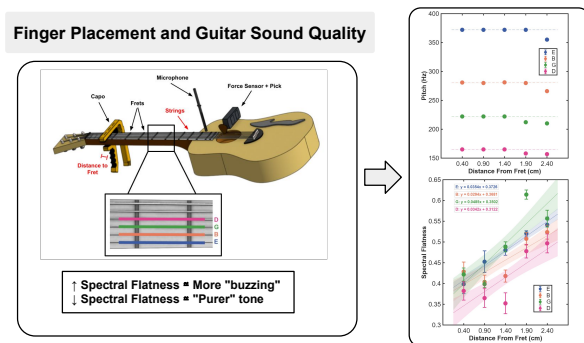


EFFECTS OF FINGER PLACEMENT ON GUITAR SOUND QUALITY

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VISUAL ABSTRACT



ABSTRACT

Guitarists are often told to press down the strings as close to the fret as possible to produce the best sounding chords. This study analyzed the relationship between finger distance to fret and sound quality across 4 strings (high E, B, G, and D) of a standard acoustic guitar. A capo was used to apply a constant force across 5 discrete distances to the second fret while the respective strings were plucked with a controlled force. Sound was recorded with a microphone and an FFT was performed to calculate the Pitch and Spectral Flatness. A consistent increase in Spectral Flatness occurred as distance from the fret increased. Pitch remained constant across the 3 closest distances but became flatter past a certain threshold. The results confirm that guitarists should indeed place their finger as close to the fret as possible, and also provide insight into which fingers are more important to place accurately.

Keywords: Guitar, Fret, Pitch, Spectral Flatness

1. INTRODUCTION

When a novice guitarist attempts to learn the instrument, one of the first pieces of advice in tutorials all over is to place their finger as close as possible to the guitar's fret—the vertical lines which press into the strings, determining the desired

pitch—when they are playing a chord, or a combination of different strings pressed down to produce a certain sound. A brief qualitative test quickly reveals the efficacy of this advice. Anyone picking up a guitar can notice that, if they locate their fingers particularly far from the desired fret or push down with particularly little amount of force, the string will produce an undesirable, distorted “buzzing” sound. However, it’s unclear where the limit is between playing a high-quality sound, and playing a distorted sound; or, whether this limit may differ across the strings of a standard acoustic guitar. Additionally, many chords require complex fingering across a variety of different strings. Often, it is impossible to locate each finger reasonably close to the fret, and as a result, sound quality may suffer. This dilemma motivates an analysis into the relationship between finger distance to fret, and sound quality, as well as how this relationship differs across strings. An in-depth and quantitative answer to this question would allow guitarists all over, and especially beginners, to understand how important finger placement is across the chords that they want to play.

This section will discuss various experiments studying different independent variables and inputs for a guitar player, as well as multiple analyses introducing different metrics to quantify and discuss sound quality. However, there remains a gap in the field for analyzing not just which string and which fret is being played, but also how small differences in finger placement between frets affects output sound.

To fill this gap in current research, and help inform beginner guitarists, a methodology is provided for studying the effects of finger placement on sound quality across the 4 highest strings of a standard guitar. 5 discrete distances were chosen to a particular fret, and 5 trials of plucking the string while using a capo (a standard device used to keep guitar strings held down when guitarists wish to change key) to press down at the respective distance were performed for each distance. This was repeated across the 4 highest strings, D, G, B, and high E. Trials were gated for quality using a force sensor attached to the pick, and only trials where the measured plucking force fell within a desired threshold were chosen for downstream data analysis. Raw audio was picked up using a

microphone positioned to accept the majority of the sound, while minimizing the picked-up external noise as much as possible. Then, a Fast Fourier Transform (FFT) was performed on the raw audio, converting from the time to the frequency domain and allowing for derivation of the sound quality metrics of pitch—the frequency at which the first harmonic occurs—and spectral flatness—a measure of how noisy vs. concentrated the frequency domain data is. Finally, the outputted relationships were fit to curves and monitored across all 4 strings to determine different relationships or boundary conditions between finger placement and sound quality by string. With these results, future generations of beginner guitarists can make educated and empirically-backed decisions on how to value or prioritize their finger placements.

2. BACKGROUND

Sound and how we produce it is a phenomenon that has been extensively studied, including looking into how the guitar in particular produces sound. This section will begin by providing a mathematical framework of sound and sound waves in general in order to motivate finding an appropriate quantitative descriptor of sound quality; then, it will review some of the specific independent variables that have been controlled in guitar-based experiments in order to position finger distance to fret as a novel control variable.

2.1. Measurements of Sound and Sound Quality

Prior research on sounds produced by plucking guitar strings has converged upon a simplistic physical model for a vibrating guitar string. The characteristics of a vibration strongly influence the corresponding sound, so understanding the mathematical framework behind these vibrations will allow for an appropriate choice of sound quality metric. The standard wave equation is used to model a particular guitar string as a flexible string with pinned boundaries [1], where the vibrations of the string are governed by the standard transverse wave equation, Equation 1, for a vibrating string:

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad (1)$$

where x represents a length along the dimension of the string, and $c^2 = \frac{T}{\mu}$ relates the string tension T and the string's linear density μ [2]. This framework allows us to understand how further aspects of sound quality will shift depending on parameters such as string length, density, and tension.

Determining "quality" of sound is in many ways a subjective procedure. However, there are a few general aims that a guitarist strives for when playing any chord. The first and most standard measurement of sound is pitch. Following the aforementioned physical model, pitch, also known as the fundamental frequency, can be mathematically determined via Equation 2 [2]:

$$f_1 = \frac{c}{2L} = \frac{1}{2L} \sqrt{\frac{T}{\mu}} \quad (2)$$

with f_1 representing the fundamental frequency, c representing the speed of light, and L representing the length of the string. This model allows pitch (fundamental frequency) to be completely determined by the parameters of string length (L), density (μ),

and tension (T). Though pitch may not be directly correlated to sound quality, when playing certain chords, a desired pitch is meant to be reached, and deviations from this desired pitch may cause chords to sound worse.

Finally, a large measure of sound quality in a guitar is how distorted or "noisy" the plucking of the string sounds. Any guitarist can qualitatively witness how, if a string is not pressed down hard enough against a fret, it will vibrate excessively and cause a noticeable distortion in the produced sound. In order to quantify this measurement, Spectral Flatness is introduced as a metric representing how much of the sound amplitude can be attributed to clean frequency peaks as opposed to noise. Introduced distortion, or inharmonicity, will shift partials to higher frequencies than desired, resulting in a lower perceived sound quality [3]. These "stretched" frequencies are determined mathematically by Equation 3 [4]:

$$f_n = n f_1 \sqrt{1 + B n^2} \quad (3)$$

where f_n represents the n th harmonic, f_1 represents the first harmonic (pitch), and B is a determined inharmonicity coefficient. To utilize this theory to describe an accessible and easily digestible format useful for answering the objective question of this particular experiment, spectral flatness is introduced as a measure of sound quality. Spectral flatness is a commonly used metric to determine how "noisy" a sound or signal is, and is calculated as the ratio of the geometric mean to the arithmetic mean of the frequency-domain sound amplitudes [5]. Spectral flatness represents how much of the overall sound is contributed from discrete peaks vs. flat noise, and hence is used as a metric for "purity" or "sharpness" of tone, with values near 0 corresponding to very "sharp" tones, and values near 1 corresponding to noisy sound [6].

2.2. Previously Studied Inputs

Previous studies have tracked guitar sound across various inputs different inputs. One commonly studied input is the motion of the plucking hand [7]. There is certainly a lot of variation in output sound that depends on the particular motion and mechanics in which the guitarist moves their hand. Variables such as angle of plucking hand and distance at which the string is plucked from the bridge have been shown to have a measurable impact on sound quality [7].

There have also been significant efforts to use computational models to estimate which string is being played from which fret purely from the outputted sound [1]. These tools use many of the previously mentioned mathematical string models to essentially reverse-engineer and predict what set of inputs was used to generate the outputted sound. While these methods have shown promising results, they fail to account for micro-displacements in between frets and instead always assume perfect string content to fret.

Despite rich theoretical background as well as practical experimentation regarding guitar string plucking, there remains a gap in the current literature. Specifically, there exists a dearth of current studies, to the best of the author's knowledge, that examine finger displacement between frets, and whether different finger placements in between two frets has an effect on the sound

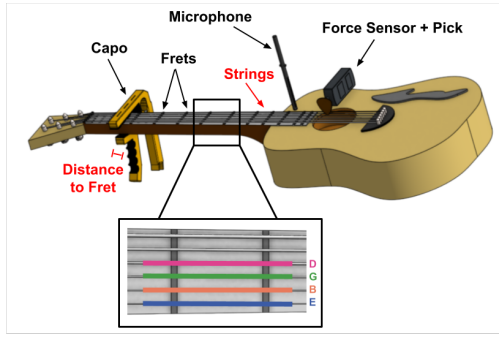


FIGURE 1: CAD diagram of the setup used to consistently measure sound while plucking strings with a capo at varied distances to the fret. Which string was plucked and the capo's distance to the fret are the independent variables, highlighted in red. Only the 4 highest strings, D, G, B, and high E, were tested.

quality of the plucked string, hence motivating the experiment outlined in this paper.

3. EXPERIMENTAL DESIGN

This experiment used a standard acoustic guitar, with a setup designed to record plucking noise at as consistent of a sound level as possible, with as little background interference as possible. A diagram of the experimental setup can be seen in Figure 1.

In the experiment, the 4 highest strings (D, G, B, and high E) were tested across 5 evenly spaced distances (0.40 cm, 0.90 cm, 1.40 cm, 1.90 cm, and 2.40 cm) between the capo and the second fret. A Vernier Microphone was hung directly above the soundhole to ensure a minimal amount of background noise was picked up, while receiving as much of the guitar's inherent sound as possible. During each trial, the capo was moved to the appropriate distance to the fret, then a guitar pick was attached to a Vernier Dual-Range Force Sensor and the appropriate string was plucked. In order to provide consistency of plucking force, only trials whose maximum plucking force fell in a threshold of $1.0 \text{ N} \pm 0.1 \text{ N}$ were kept. For each data point (in total, 4 strings $\times$ 5 distances = 20 data points), 5 trials which fell within the desired force threshold were kept.

4. RESULTS AND DISCUSSION

4.1. Converting to the Frequency Domain

Each of the four strings was plucked with a consistent force across the varying distances to the second fret, and the raw sound was measured using a microphone held at a close and consistent distance to the strings. Example data of a raw audio recording is shown in Figure 2.

While it is possible to notice repeating sinusoidal sections from this raw amplitude vs. time graph, a frequency domain analysis is useful for determining important sound properties, such as pitch and the frequencies at which different harmonics occur. In order to analyze this sound data in the frequency domain, a Fast Fourier Transform (FFT) following the standard Discrete Fourier

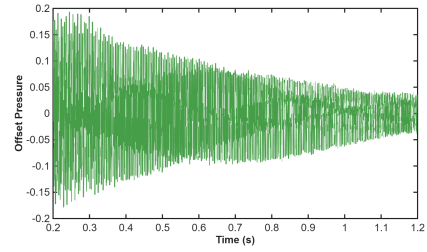


FIGURE 2: Sample raw audio data, from the G string, 0.40 cm from the second fret, trial 1. Offset Pressure is measured over time from a microphone located close to the string being plucked. A stable portion—beyond the “attack”, or initial amplitude spike, but before the amplitude gets too low—of the waveform (in this case, 0.2 seconds $\rightarrow$ 0.5 seconds), is chosen for downstream analysis, including an FFT. Frequency domain analysis was performed to determine the metrics of pitch and spectral flatness.

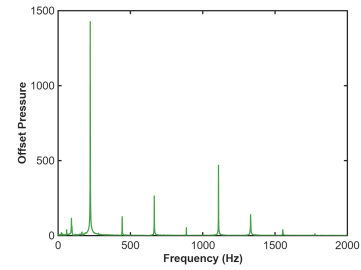


FIGURE 3: Sample frequency domain audio data, resulting from performing an FFT on the same raw audio data shown in Figure 2 (G string, 0.40 cm from the second fret, trial 1). Each harmonic can clearly be seen as the peaks. In this case, it can be seen that the pitch is represented by the fundamental frequency, or the tallest peak at around 250 Hz, in between 0 Hz and 500 Hz. Each following harmonic has a frequency at an integer multiple of the fundamental. Frequency domain data which is more concentrated in discrete peaks will correspond to a “purer” sound than frequency domain data that is spread out or noisy.

Transform (DFT) equation shown in Equation 4 was used:

$$X(k) = \sum_{n=1}^{N-1} |x(n)| e^{\frac{-j2\pi kn}{N}} \quad (4)$$

Figure 3 shows the resulting frequency domain plot, after FFT has been performed on the same data from Figure 2.

Expanding upon this, frequency domain data was computed across all strings, distances, and trials using FFT. The data being in the frequency domain rather than the time domain allows for much more meaningful observations to be made. As a baseline for this, the mean Offset Pressure vs. Frequency graph was computed across all 5 trials for all 5 distances to the fret. Figure 4 shows the mean frequency domain graph for each of the 5 distances. It can be observed how for distances closer to the fret, the first peak (at around 250 Hz in Figure 3), or fundamental frequency, dominates. Additionally, the Offset Pressure axis has a much larger range, showing that more of the sound can be attributed to the fundamental frequency. This generally correlates to a “purer” sound, without overtones from the other harmonics interfering with the tone of the first harmonic. Similarly,

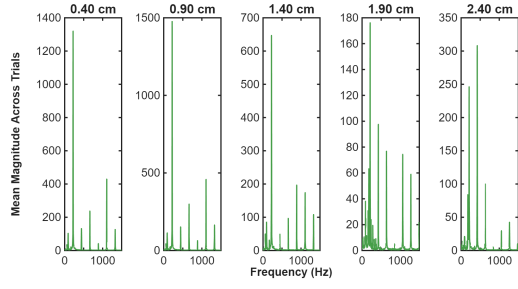


FIGURE 4: Mean magnitude of Offset Pressure across all 5 trials for 5 distances to the fret, on the G string. It can be observed that as distance to the fret increases, the frequency domain peaks become more evenly distributed and the amplitude of the first harmonic decreases, showing a more chaotic and less “pure” tone.

for distances farther from the fret, the Offset Pressure peaks are more evenly distributed, which generally correlates to a “noisier”, “buzzier” sound that is less “pure” in tone.

4.2. Pitch

From the frequency domain data, more important and useful metrics can be obtained. First, pitch can be easily computed as the peak at the lowest frequency. Pitch is also referred to as the fundamental frequency, or the frequency at which the first harmonic appears. Computationally, pitch was determined from the frequency domain data by searching for the highest amplitude peak in a frequency window determined by existing knowledge of a reasonable range in which the given note’s pitch would occur, represented in Equation 5:

$$f_1 = \arg \max_{f \in [0.8f_{\text{exp}}, 1.2f_{\text{exp}}]} |X(f)| \quad (5)$$

Figure 5 shows the final plot of mean pitch across all 5 distances to the fret on all 4 strings.¹ It has been shown that as long as the capo was at least 1.40 cm from the fret, all of the strings consistently produced the desired pitch. However, the E and B strings exhibited a decrease in pitch starting at 2.40 cm from the fret, and the G and D strings exhibited a decrease in pitch starting at 1.90 cm from the fret. These results may motivate guitarists to prioritize keeping their fingers close to the fret for the G and D strings more so than for the E and B strings when playing complex chords where certain arrangements of fingers may be difficult or impossible.

4.3. Spectral Flatness

While pitch is an important metric for determining how we hear a sound and what note is being heard, it fails to encapsulate how pleasing a sound is, or sound quality. In order to determine how sound quality changes with distance to the fret, spectral flatness was calculated based on the frequency domain graphs. Spectral flatness measures how “pure” a tone is; a string that’s constantly vibrating against the fret because it’s not making clean contact will produce a generally unpleasant buzzing noise.

¹95% error bars are technically included in the script to produce this plot, but don’t appear as visible because the pitch calculations were exactly the same across trials, leaving no error at all.

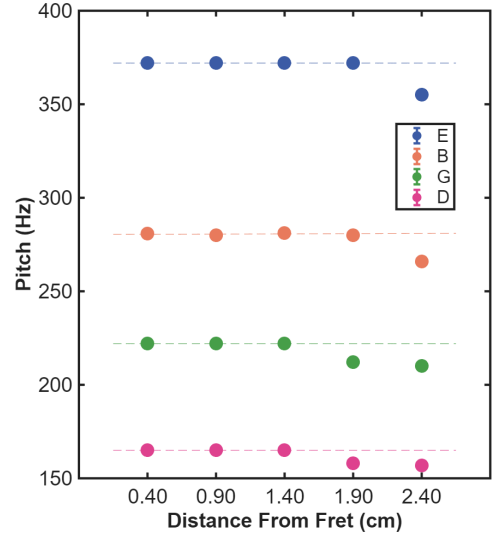


FIGURE 5: Pitch vs. Distance From Fret for all 4 strings, across all 5 distances. Each data point represents the mean of each pitch calculation. 95% error bars are technically included, however because the pitch calculation returned the exact same pitch value for each string at each distance to the fret, the variance is 0 and the error bars do not appear on the plot. The dotted lines represent the pitch of the closest 3 distances to the fret, and hence are the desired pitch to be obtained when pressing down on the relevant string at the second fret. Across the 3 closest distances to the fret, the pitch stays constant for each string. For the E and B strings, the pitch is still constant at 1.90 cm from the fret, then decreases at 2.40 cm from the fret. For the G and D strings, the pitch decreases at 1.90 cm from the fret. Physically, these decreases in pitch imply that the string is no longer making direct contact with the fret, so a different note than intended is being played.

Spectral flatness was calculated from the frequency domain data as the ratio between the geometric mean and the arithmetic mean of the non-zero magnitudes as shown in Equation 6:

$$SF = \frac{\exp\left(\frac{1}{N} \sum_{k=1}^N \log |X(f_k)|\right)}{\frac{1}{N} \sum_{k=1}^N |X(f_k)|} \quad (6)$$

By this formulation, spectral flatness ranges from 0 to 1, with values close to 0 representing an extremely pure tone (most of the amplitude is from the harmonic peaks) and values close to 1 shifting towards noise (flatter spectrum).

Spectral flatness was computed for all data points, and the mean and 95% bounds were calculated across the 5 trials. Figure 6 shows the resulting spectral flatness data across finger distance to fret and string. All 4 strings exhibited a statistically significant linear upwards trend in spectral flatness as distance to the fret increased, aligning with the physical intuition: as the finger or capo moves farther from the fret, the contact between the string and fret becomes less secure, causing the string to vibrate against the fret and leading to the “buzzing” sound which correlates to the increase in spectral flatness. The G string showed the steepest change in spectral flatness as distance increased, while the B string showed the least steep change. These results prove the intuition that to maintain the purest tone possible and the highest sound quality, guitarists should indeed attempt to press their fingers as close to the fret as possible.

4.4. Application

Oftentimes, however, it’s not possible to keep each finger right next to the fret, especially when playing more complex chords, where the natural biomechanics of our fingers may not allow us to place them all right next to the fret. In such cases, guitarists may have to prioritize which strings they will be able to press close to the fret, and which strings they might have to settle with pressing farther away. Based on the results of this experiment, in order to play the correct note, guitarists must make sure to press less than 2.40 cm from the fret for the high E and B strings, and less than 1.90 cm from the fret for the G and D strings (Figure 5). Additionally, guitarists who wish to maximize sound quality and purity of tone should prioritize keeping their finger as close to the fret as possible for the G string, followed by the E and D strings, and may put less weight on finger distance to fret for the B string (Figure 6). An analysis of a few common chords and recommendations for how to play them based on the quantitative results of this study are shown in subsection A.1.

This experiment has successfully filled the gap in research into how finger distance to a fret affects guitar sound, providing a quantitative mathematical framework and 4 individual mathematical relationships across the 4 highest guitar strings for how both pitch and sound quality (measured via spectral flatness) change with finger distance to fret. It utilized existing, robust measures of sound (pitch and spectral flatness) with a novel independent variable (distance to fret) to expand upon existing research into how playing the guitar in different ways can affect the resulting sound.

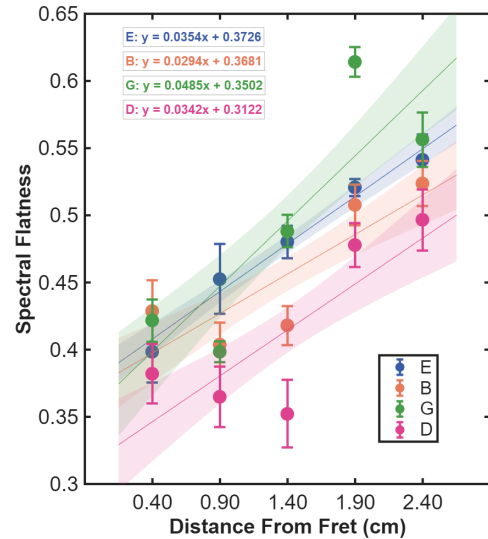


FIGURE 6: Spectral flatness vs. Distance From Fret for all 4 strings, across all 5 distances. Each data point represents the mean, with 95% error bounds, of the spectral flatness across 5 trials. Each string was fit to a linear model of the form $y = ax + b$, and 95% confidence bounds on the fit are shaded. The calculated fits, including the slope and intercept, are shown in the top-left corner. They are all statistically significant. Each strings exhibited a general increase in spectral flatness as distance to the fret increased, aligning with the expectation that fingers farther from the fret tend to produce more vibrations and buzzing (corresponding to higher spectral flatness). The G string exhibited the steepest increase in spectral flatness as distance increased, showing sound from plucking it may be more sensitive to changes in figure distance. The B string exhibited the least steep increase in spectral flatness as distance increased, showing sound from plucking it may be less sensitive to changes in finger distance.

5. CONCLUSION

In order to validate the accuracy of the commonplace advice of pressing the strings as close as possible to the fret when playing guitar, and to provide a mathematical framework for the relationship between finger distance to fret and sound quality, an experiment was set up in which sound was recorded from plucking each of the 4 highest strings on a standard acoustic guitar at 5 discrete distances to the second fret. Frequency domain analysis was done on the resulting audio recordings, and the metrics of pitch and spectral flatness were computed across trials.

It was found that pitch stayed consistent up to 1.90 cm to the fret for the high E and B strings, after which it decreased. For the G and D strings, pitch stayed consistent up to 1.40 cm to the fret, after which it decreased.

Spectral flatness was used as a measure of sound quality, describing the “purity” of the recorded tone. Spectral flatness was found to increase at a statistically significant linear rate as the distance to the fret increased for all 4 tested strings. The G string exhibited the most sensitivity to distance from the fret, and the B string exhibited the lowest sensitivity to distance from the fret. Considering the G string also showed a decrease in pitch at 1.90 cm, it can be concluded that the G string is particularly sensitive to how far from the fret it is being pressed, and guitarists should strive to prioritize pressing it as close as possible. On the other hand, considering the B string only started decreasing in pitch at 2.40 cm, it can be concluded that the B string is the least sensitive of the 4 tested strings, and guitarists will still be able to produce a good quality sound even if their finger is pressing the B string a little farther from the fret.

The above results provided a quantitative framework for how changing a guitarists’ finger location when plucking strings changes the resulting sound quality, allowing guitarists to make mathematically informed decisions regarding which strings they should prioritize pressing down close to the fret when fingering complex chords. The experiment did also raise further questions and opened the door for a deeper analysis into testing how different finger placements or orientations can affect sound quality. Future work should include the 2 other strings of a standard 6-string acoustic guitar in order to investigate whether they share the same patterns. Additionally, varying not only the distance to the fret but also the force with which the string is pressed down could further elucidate how different fingering styles could affect guitar’s sound. Finally, other metrics for sound quality could be explored, including taking a more detailed look at individual integer harmonics and seeing how they might differ as finger distance from the fret changes.

ACKNOWLEDGMENTS

The author would like to thank Dr. Lachner and Professor DiGenova for their assistance, feedback, and support.

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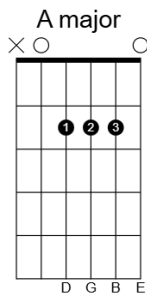
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APPENDIX A. CHORD FINGERING RECCOMENDATIONS BASED ON EXPERIMENTAL RESULTS

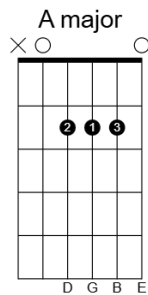
The following are some common chords, and recommendations for how to play them based on the quantitative results from this study. The results for both pitch and spectral flatness as a function of finger distance to the fret have been taken into account, as well as a basic understanding of common fingering patterns, to make these recommendations.

A.1. A Major

There are two common ways to play the A major open chord:



Version 1

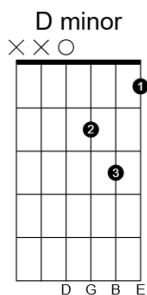


Version 2

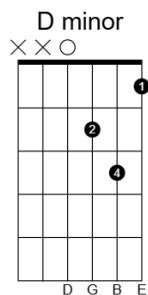
Due to the nature of how our hands are shaped, finger 1 (index finger) will generally be the furthest from the fret. Based on the findings of this paper, namely that the G string is the *most* sensitive to distance to fret and the B string is the *least* sensitive, it would make sense to prioritize pressing the G string as close as possible to the fret. Hence, Version 1 in the above image is recommended, as it allows the G string to be pressed closer to the fret (with finger 2, the middle finger).

A.2. D minor

The D minor chord can be difficult to play well, due to the fact that there's a large stretch over 3 different frets. Thus, some guitarists choose to press the B string with finger 4 (pinky) instead of finger 3 (ring finger):



Version 1



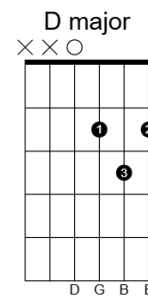
Version 2

The results of this paper showed that, even though the B string was indeed the least sensitive to changes in finger distance to the fret, it still decreased in sound quality as the finger distance to the fret increased, and still dropped off in pitch when the finger

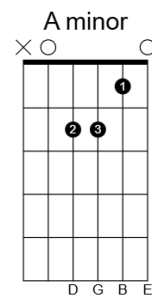
distance to the fret was particularly far. Hence, guitarists who find that they are unable to stretch enough and can not reach close enough to the third fret when playing Version 1 should consider switching to Version 2, where they will have to stretch less and will be able to more comfortably play the B string close to the third fret.

A.3. D major and A minor

D major and A minor are both very popular chords. While there's really only one common way to finger both of these chords, it is notably that they both involve pressing down on the G string. The above results show just how sensitive the G string can be to changes in finger distance to fret affecting sound quality, so guitarists should take particular effort, when playing these two chords, to keep the G string finger (finger 1 for D major, finger 3 for A minor) as close to the fret as possible, even if it may be at the expense of letting some of the other fingers move slightly further from the fret.



D major



A minor